

N100 FINANCIAL INTELLIGENCE PLATFORM

Comprehensive Institutional Analyst User Guide & System Manual
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Target Audience:	Equity Research Analysts, Portfolio Managers, Data Engineers
Coverage Universe:	92 NIFTY 100 Companies across 11 Sector Groups
REST API URL:	http://127.0.0.1:8000/api/v1
Dashboard Port:	http://127.0.0.1:8501 (Streamlit)
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1. Platform Architecture & Data Pipeline

The N100 Financial Intelligence Platform is engineered as a end-to-end data analytics and reporting system for institutional equity research. It ingests multi-year raw financial statements, market data, corporate actions, and company metadata across 92 companies, normalises schema representations, evaluates 16 rigorous Data Quality (DQ) validation rules, computes 40+ financial ratios, runs machine learning clustering, and exposes data through both a Streamlit dashboard and a FastAPI REST service.

Data Flow Architecture:

Phase 1: ETL & Schema Normalisation	Ingests Excel/CSV statements, standardises ticker formats, normalises years, and enforces schema strictness.
Phase 2: Validation Engine (DQ Rules)	Evaluates 16 validation rules (DQ-01 to DQ-16) covering primary keys, referential integrity, and logical checks.
Phase 3: Database Storage (SQLite)	Stores normalised relational records across 10 core tables in data/db/nifty100.db with foreign key constraints enabled.
Phase 4: Financial & ML Analytics Engine	Computes DuPont 3/5-step decompositions, CAGR metrics, CFO quality scores, and 5-cluster KMeans archetypes.
Phase 5: Presentation Layer	Multi-page Streamlit Dashboard (Port 8501) + FastAPI REST Web Service (Port 8000) + ReportLab PDF Generator.

Key System Files & Paths:

- **Database File:** data/db/nifty100.db
- **FastAPI Server Scaffold:** src/api/main.py
- **Clustering Analytics:** src/analytics/clustering.py
- **DuPont Decomposition:** src/analytics/dupont.py
- **Tearsheet PDF Generator:** src/reports/tearsheet.py
- **Streamlit Main Entry:** app.py

2. Streamlit Dashboard Navigation & Workflows

The Streamlit dashboard provides an intuitive, web-based UI for analysts to inspect portfolio trends, perform deep-dive company analysis, run multi-factor screeners, and generate PDF tearsheets.

Dashboard Screen Matrix:

Screen Name	Path	Key Capabilities
01 Executive Overview	pages/01_overview.py	Macro portfolio KPIs, sector weightings, top market-cap leaders.
02 Company Profile	pages/02_profile.py	Deep-dive financial profile, 10-yr P&L, BS, CF history, and DuPont breakdown.
03 Multi-Factor Screener	pages/03_screener.py	Custom metric sliders, preset quality screeners, CSV data export.
04 Peer Comparison	pages/04_peers.py	8-axis radar comparison charts, percentile ranks within 11 peer groups.
05 Financial Trends	pages/05_trends.py	YoY growth charts, margin expansion trends, capital expenditure analysis.
06 Sector Intelligence	pages/06_sectors.py	Sector medians, ROE vs P/E scatter plot, sector concentration index.
07 Capital Allocation	pages/07_capital.py	CFO quality scores, FCF conversion rate, leverage risk classification.
08 Reports & Export	pages/08_reports.py	Instant PDF tearsheet generator, batch PDF download, data export.

How to Start the Dashboard:

Execute the following terminal command from the workspace root:

```
streamlit run app.py
```

3. Interactive Stock Screener & Filtering Logic

The Multi-Factor Stock Screener enables analysts to filter 92 companies simultaneously across valuation, profitability, growth, and leverage dimensions.

Screener Filter Parameters:

Parameter	Range / Type	Description & Purpose
Minimum ROE (%)	-50% to +100%	Filters out companies below target Return on Equity threshold.
Maximum D/E Ratio	0.0 to 10.0	Filters out over-leveraged companies (e.g. D/E > 1.5).
Min 5-Yr Rev CAGR	-20% to +50%	Requires minimum 5-year top-line growth rate.
Min 5-Yr PAT CAGR	-20% to +50%	Requires minimum 5-year bottom-line growth rate.
Maximum P/E Ratio	0.0 to 200.0	Filters out expensive valuation multiples.
Minimum FCF (Cr)	-1000 to +10000	Ensures positive free cash flow generation.
Sector Filter	Select from 11 Sectors	Restricts results to specific broad sector or industry.

Preset Quality Screener:

The preset 'High Quality Compounders' preset applies standard institutional filters:

- ROE >= 15% AND Debt-to-Equity <= 1.0 AND 5-Yr Rev CAGR >= 10% AND FCF > 0.

This preset typically yields between 10 and 50 top-tier NIFTY 100 companies.

4. Institutional PDF Tearsheet Generation

The platform generates 2-page publication-quality PDF tearsheets for each company in the NIFTY 100 universe. These tearsheets adhere to strict layout boundaries (zero text overflow, high-contrast tables, and dynamic headers).

Tearsheet Content Breakdown:

Page 1 Elements	Page 2 Elements
• Company Header & Sector Metadata • Latest Valuation & Profitability KPIs • 5-Year Financial Summary Table • DuPont 3-Step & 5-Step Breakdown Table • NLP-Generated 3 Pros & 3 Cons	• 10-Year P&L; Statement Summary • 10-Year Balance Sheet Summary • Capital Allocation & FCF Quality Score • Peer Group Ranking Table • Analyst Recommendation & Target Price

CLI Command for Batch Tearsheet Generation:

```
python -c "from src.reports.tearsheet import batch_generate_tearsheets;
batch_generate_tearsheets()"
```

5. Machine Learning KMeans Clustering & Archetypes

To group companies into objective financial behavioral profiles, the platform implements KMeans clustering with 5 clusters operating on 5 core metrics.

Cluster Feature Set & Preprocessing:

- 1. **Feature Metrics:** return_on_equity_pct, debt_to_equity, revenue_cagr_5yr, fcf_cagr_5yr, operating_profit_margin_pct.
- 2. **Missing Value Imputation:** Replaced with sector median for each metric prior to scaling.
- 3. **Normalisation:** StandardScaler converts all features to zero mean and unit variance.
- 4. **Reproducibility:** KMeans(n_clusters=5, random_state=42).

5 Cluster Archetypes:

Cluster ID	Archetype Name	Financial Characteristics
0	High-Quality Compounders	High ROE (>20%), strong FCF CAGR, low debt-to-equity (<0.5).
1	Defensive Dividend Payers	Stable OPM, high dividend payout, moderate growth, low volatility.
2	Value Cyclicals	Moderate P/E ratio, capital intensive, cyclical top-line growth.
3	Distressed or Turnaround	High leverage (D/E > 2.0), negative/low FCF CAGR, erratic ROE.
4	Emerging Growth	High revenue CAGR (>15%), reinvesting cash, expanding margins.

6. FastAPI REST Service & Endpoint Reference

The FastAPI server exposes 16 REST endpoints with CORS enabled and request duration logging. Server runs on `http://127.0.0.1:8000` with OpenAPI docs at `/docs`.

16 REST API Endpoints Summary:

Method	Endpoint Path	Description
GET	<code>/api/v1/health</code>	Returns API status, DB table row counts, uptime, version.
GET	<code>/api/v1/companies</code>	List all 92 companies with sector, roe_pct, roce_pct.
GET	<code>/api/v1/companies/{ticker}</code>	Full profile, latest KPIs, sector data (404 if invalid).
GET	<code>/api/v1/companies/{ticker}/pl</code>	P&L; history array (from_year, to_year filters).
GET	<code>/api/v1/companies/{ticker}/bs</code>	Balance sheet history array.
GET	<code>/api/v1/companies/{ticker}/cashflow</code>	Cash flow statement history array.
GET	<code>/api/v1/companies/{ticker}/ratios</code>	Computed financial ratios per year.
GET	<code>/api/v1/companies/{ticker}/tearsheet</code>	Binary PDF tearsheet download (application/pdf).
GET	<code>/api/v1/screener</code>	Multi-factor screener with min_roe, max_de, max_pe filters.
GET	<code>/api/v1/sectors</code>	Returns 11 sectors with median ROE, PE, DE metrics.
GET	<code>/api/v1/sectors/{sector}/companies</code>	Companies within specified sector.
GET	<code>/api/v1/peers/{group_name}</code>	Companies in peer group with percentile ranks.
GET	<code>/api/v1/companies/{ticker}/peers/compare</code>	Radar chart data (8 axes for company + peer average).
GET	<code>/api/v1/market-cap/{ticker}</code>	Valuation multiples history (P/E, P/B, EV/EBITDA) 2019-2024.
GET	<code>/api/v1/portfolio/stats</code>	P10 to P90 percentile table across all 92 companies.
GET	<code>/api/v1/companies/{ticker}/documents</code>	Annual report links with is_url_valid boolean flag.

Example curl Command:

```
curl -X GET "http://127.0.0.1:8000/api/v1/screener?min_roe=15&max_de=1.0"
```

7. Data Quality & Validation Failure Tracking

Data Quality is strictly enforced by 16 validation rules evaluating inputs before database insertion. Failures are logged to output/validation_failures.csv and the validation_failures SQLite table.

16 Data Quality Rules Summary:

Rule ID	Severity	Rule Description & Constraint
DQ-01	CRITICAL	Primary key uniqueness check (no duplicate tickers/years).
DQ-02	CRITICAL	Primary key non-null and valid year format [2000, 2030].
DQ-03	CRITICAL	Referential integrity check (child records must exist in companies master).
DQ-04	WARNING	Balance Sheet accounting identity (Assets == Liabilities + Equity within 1%).
DQ-05	WARNING	OPM cross-check (reported OPM vs computed OP/Sales within 5%).
DQ-06	WARNING	Positive revenue/sales constraint (Sales > 0).
DQ-07	WARNING	Cash Flow reconciliation (Ending Cash == Start Cash + Net Cash Flow).
DQ-08	WARNING	Tax rate sanity check (Effective tax rate between 0% and 100%).
DQ-09	CRITICAL/WARNING	Positive stock prices (>0) and dividend payout cap (<= 500).
DQ-10	WARNING	Company website URL format regex validation.
DQ-11	WARNING	EPS sign consistency (EPS and Net Income must share same sign).
DQ-12	WARNING	Exchange ticker suffix check (.NS, .BO, .BSE, .NSE).
DQ-13	WARNING	Minimum price record count coverage & high leverage warnings.
DQ-14	WARNING	Stock daily trading volume non-negative check (Volume >= 0).
DQ-15	WARNING	Daily price consistency (High >= Open/Close/Low and Low <= Open/Close).
DQ-16	WARNING	Logical profit order (Gross Profit >= Operating Profit >= Net Income).

8. Performance Optimization & Benchmark Guidelines

To guarantee responsive user experience across both Streamlit dashboard and FastAPI endpoints, database and execution optimization strategies were implemented.

Performance Benchmarks Summary:

Metric / Benchmark	Target Threshold	Achieved Performance	Status
Concurrent API Screener Calls (10)	< 10.0 seconds	0.125 seconds	PASS
Company Profile Screen Load Time	< 3.0 seconds	0.007 seconds	PASS
Full PyTest Test Suite Execution	< 30.0 seconds	18.60 seconds (245 tests)	PASS
Batch Tearsheet PDF Generation (92)	< 60.0 seconds	12.40 seconds	PASS

SQLite Optimization Best Practices Applied:

- Write-Ahead Logging (WAL):** PRAGMA journal_mode=WAL; enabled to allow non-blocking concurrent reads.
- Compound Indexing:** Compound index (ticker, year) on large tables (financial_ratios, profitandloss, balancesheet, cashflow).
- Connection Reuse:** Shared thread-local connection pool in DatabaseManager.
- Memory Caching:** PRAGMA cache_size = -64000; allocates 64MB cache.

9. Automated Test Suite & Quality Gates

The platform maintains 245 automated unit and integration tests across 5 major test packages. HTML report is generated at `reports/pytest_report.html`.

Test Package Breakdown:

Test Module / Directory	Test Count	Coverage Area
tests/etl/	105 Tests	Loader row counts, normalise_year(), ticker normalisation, validator rules.
tests/kpi/	52 Tests	ROE (+/- equity), D/E ratios, ICR, CAGR flags, CFO quality scores.
tests/dq/	14 Tests	Data Quality rules DQ-01 through DQ-16 synthetic failure verification.
tests/api/	13 Tests	FastAPI endpoints (/health, /companies, /screener, /sectors).
tests/ (Root Analytics & Integration)	61 Tests	DuPont 3/5-step, Multi-Factor screener, Monte Carlo, PDF generators, QA.

10. System Troubleshooting & Maintenance Guide

Common Issues & Fixes:

- **Issue:** SQLite database locked error.
Solution: Ensure WAL mode is active: `python -c "from src.database import DatabaseManager; DatabaseManager().get_connection().execute('PRAGMA journal_mode=WAL;')"`.
- **Issue:** Tearsheet PDF font or layout overflow.
Solution: Re-run ReportLab generator: `python -c "from src.reports.tearsheet import batch_generate_tearsheets; batch_generate_tearsheets()"`.
- **Issue:** Port 8000 or 8501 already in use.
Solution: Kill conflicting python process or specify alternate port: `uvicorn src.api.main:app --port 8001`.